



MIA CARE

Connected Medical Devices for Patient Engagement

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Connected Medical Devices: What the future reserves?

Digital innovation is revolutionizing the organization of the healthcare world, enabling new models of assistance, therapy, and care that promote an optimization of healthcare data management, a direct and immediate doctor-patient communication, and an active participation of the person in the treatment path.

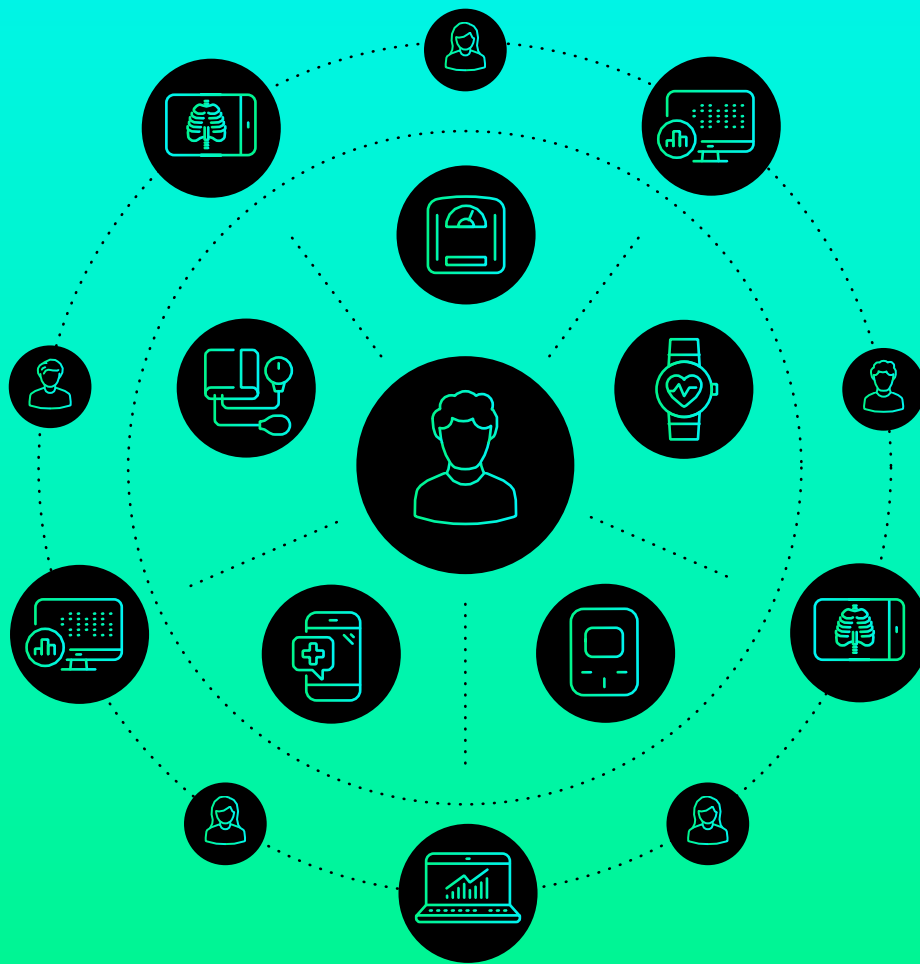
In particular, the **Internet of Things** applied to the Healthcare sector (Internet of Healthcare Things or Internet of Medical Things) enables the connection of medical devices to the digital world.



In these cases, we speak of **Connected Medical Devices**, built to improve the quality of care by assuring immediate data exchange and streamlining the resources available.

On the one hand, the connected devices allow medical staff to easily manage and monitor adherence to the prescribed therapy, offering the patient an intuitive user experience and a personalised therapeutic path. On the other hand, through the collection of accurate and reliable health data, they allow MedTech companies to verify the clinical effectiveness of the device and consequently orient the future product innovation strategy.

Connected medical devices support the **patient engagement** process, defined as a patient's active involvement in therapy and care path from a cognitive, emotional and behavioral point of view, and also the **patient empowerment**, intended as the need to provide the person with tools and knowledge to make informed decisions about one's health.



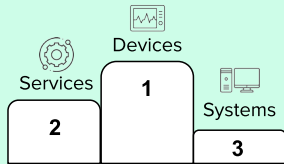
In this scenario, Mia-Care has created a **cloud-native digital platform tailored for MedTech**, which enhances the interoperability between information systems and medical devices, significantly increasing the quality of treatment and improving adherence to therapy for the well-being of patients.

As will be seen in detail in the course of the paper, the **device integration with a cloud-based platform** enables MedTechs to increase the device's intelligence, resources, and capabilities, providing increasingly advanced and personalized services.

In addition to a set of ready-to-use microservices, Mia-Care Platform permits to connect new digital applications to medical devices intuitively and rapidly, enabling the **scalability and evolution of services** based on business strategy. Meanwhile, the MedTech company can focus on the development and innovation of medical devices.

All these benefits while keeping low both new service's development and delivery costs, thanks to the technological excellence of the software stack.

Attractive Opportunities in the IoT Healthcare Market



By component, the Devices segment accounted for the highest market share of 16,6% and is anticipated to grow at the fastest rate during the forecast period.



By region, North America was the major shareholder and accounted for the highest share of \$55,272 million.

113,75 USD Billion
2019



CAGR of
13.20%

332,68 USD Billion
2027



The Global IoT Healthcare Market is expected to grow from USD 113,75 billion in 2019 to USD 332,68 billion in 2027, at a CAGR of 13.20% during the forecasted period.



Patient monitoring segment accounted for the largest growth, which is attributed to remote patient management and telehealth services.



The growth of device segment is led by wearable medical devices, increasing the trend of self-monitoring and need for analysis of the healthcare data

Source: Allied Market Research (Internet of Things in Healthcare Market)



Long-distance connectivity is transforming the MedTech sector, allowing the maximum democratization of therapeutic management, from access to care monitoring.

Patients can access digital health services remotely and take advantage of connected medical devices provided by MedTech companies based in other countries or even different continents.

2.

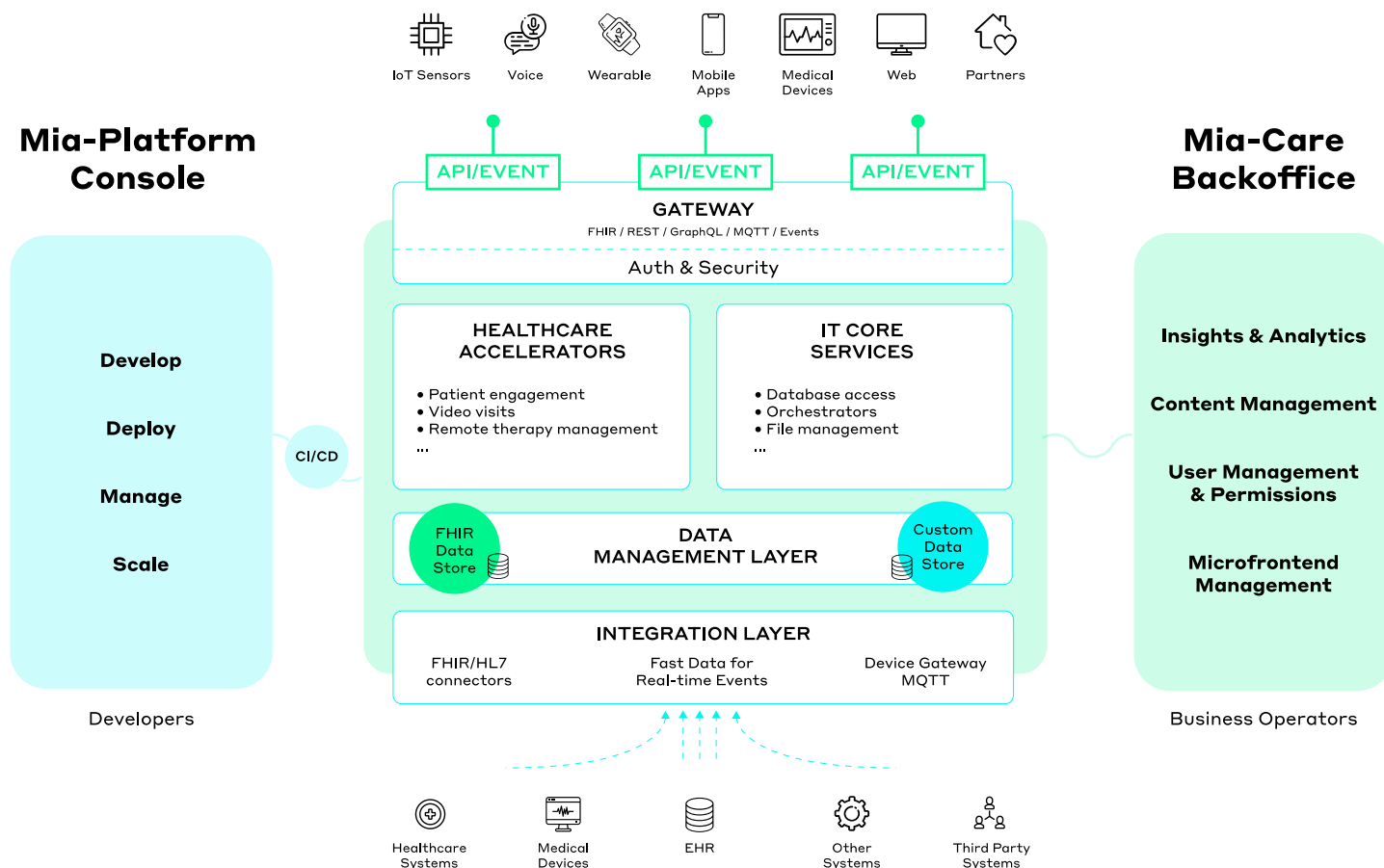
Enhance the Value-Based Health with Mia-Care, the cloud-native platform to lead the digital innovation in Healthcare

Mia-Care is a digital platform based on an **innovative technological stack that allows to easily design, create, deploy and orchestrate new digital services and products**, using a state-of-the-art architecture based on Fast Data, process automation, and service containerization. Furthermore, the highest degree of safety is ensured for all development components, complying with internationally recognized quality standards.



Mia-Care leverages the cutting-edge Mia-Platform Technology as the answer to the **need for digitalization in the healthcare sector.**

MULTIEXPERIENCE APPLICATIONS



SYSTEM OF RECORD APPLICATIONS

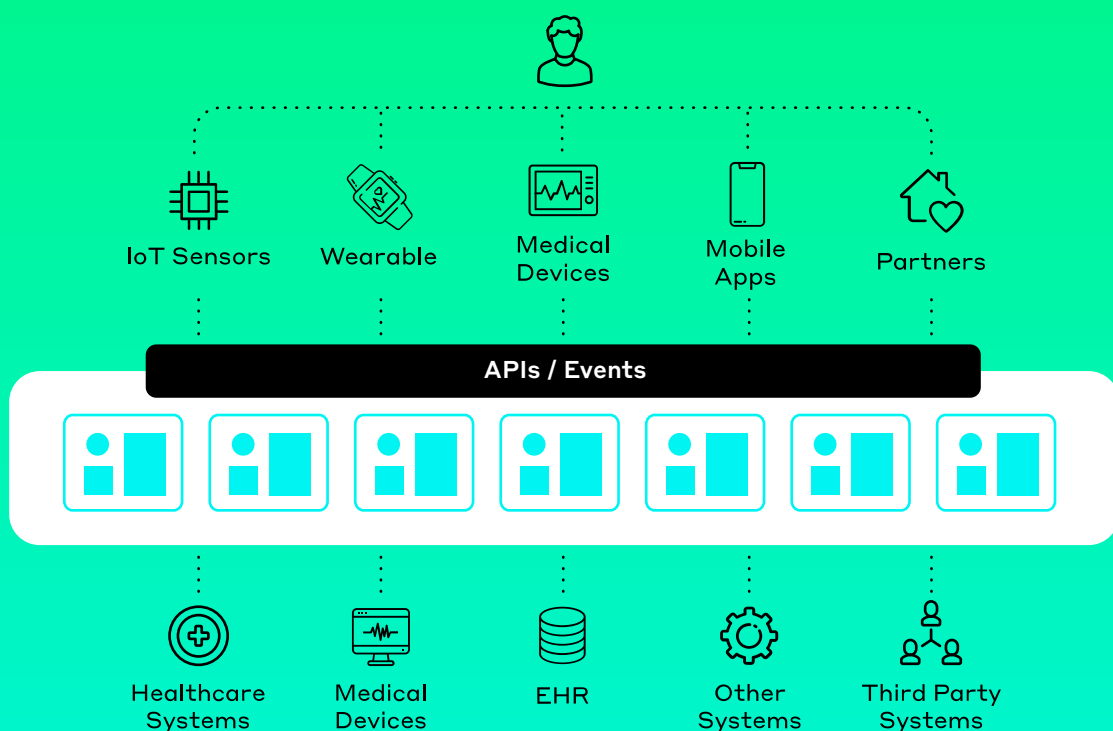
Latest-generation technologies and a cloud-native approach characterize the digital platform of Mia-Care. Thus, Feature Teams will be allowed to develop digital applications, focusing on functional logics, without the cost and complexity of autonomously purchasing, configuring, and maintaining all hardware and software components.

2.1

Fast Data Technology: collect data from connected medical devices to monitor performance in real-time and improve the therapy

Big Data is usually seen as the data collection and analysis model characterized by a level of volumes, speed, and quality that traditional technologies cannot assimilate. However, digitization has led to the need to **gather and analyze data in real-time**, an activity that collides with the computational limitations of the Big Data model. From this need arises the integration of legacy systems with Fast Data, the technology created to **acquire and monitor data immediately and flawless**.

Thanks to this innovative solution, it is possible to design and configure the complete data flow from IoT systems, creating a double value. Whereas the MedTech company can exploit the information collected to **continuously improve the device's performance** and, as a result, the quality of the service provided, medical specialists, hospitals, and caregivers have reliable information, continuously updated, and available in real-time to **address the right therapy on time**.



2.2

API Management and Gateway: quickly and easily integrate the digital channels of your healthcare ecosystem

APIs are digital intermediary components that allow the interaction of two applications without necessarily communicating with a common language and data model.

 Mia-Care enables health companies to **create and control APIs intuitively with light and performing micro-gateway.**

As a result, Development Teams will follow the entire lifecycle, from deploy to securitization, using a single access point capable of handling all the requests dynamically and quickly.

It is easy to understand the value of this layer as the trigger for the **creation of strategic partnerships**, without significantly changing your information architecture and creating interaction easy to manage with external touchpoints.

Within the Healthcare IoT landscape is **fundamental to create a flawless integration and enhance the interoperability between medical devices and the connected applications system.** Indeed, without a secure, stable, and flexible API system, it would be difficult to generate value exclusively by collecting data from the medical equipment used to manage patients' therapy. It becomes mandatory to favor a comprehensive and seamlessly exchange of information among the IT legacy system, aiming at the optimization of available resources.

2.3

Mia-Care Backoffice: manage and organize the available data in an intuitive way without the use of development code

MedTech companies can leverage a fully configurable Backoffice to easily manage multi-tenant configuration, user registration, authorization, and permissions.

Mia-Care Backoffice manages contents with a Headless CMS, provides front-end for data visualization, and finally, allows you to build new custom front-ends for different use cases.

For the purpose described in this Paper, it is important to focus on the Headless CMS. This technological component, characterized by an interface independent from the front-end, was designed to **support non-technical figures who need to manage technical elements**, such as a website or a monitoring dashboard, without editing the code.

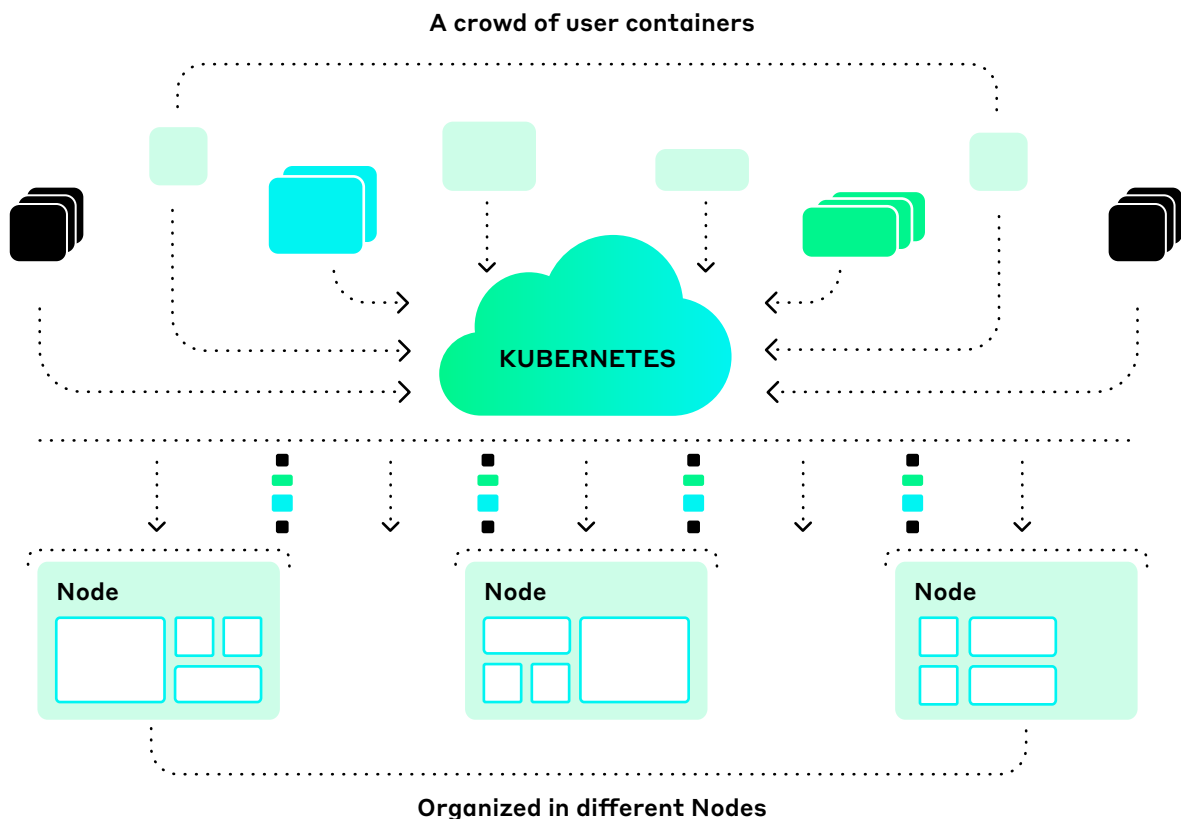
Through a digital Content Management System, both the Administration function and Hospitals General Managers can edit the visual disposition of data according to their needs, managing to achieve a **high degree of capillarity of the information collected**, patient by patient, and among structures.

2.4

Microservices Orchestration on Kubernetes: take advantage of the availability of ready-to-use and fully customizable purpose applications

Mia-Care provides an ecosystem of ready-to-use applications for the development and orchestration of your microservices.

The whole **development lifecycle of your digital products is fast and intuitive**, thanks to the rich catalog of plugins accessible through the Marketplace. The digital Accelerators developed by Mia-Care are the result of a continuous dialogue with the early adopters that helped to translate both business and functional requirements into reliable product components, customizable on different needs. As a result, the Feature Teams will only have to integrate the specifications within the microservice framework and then **deploy the business application just in a few clicks** on the selected external channels.





Develop Fast

Continuous and fluid development experience, through the automation of Operations and the orchestration strategy set on Kubernetes



Develop Smart

No lock-in effect, as all the code created lands on the client's Git repository



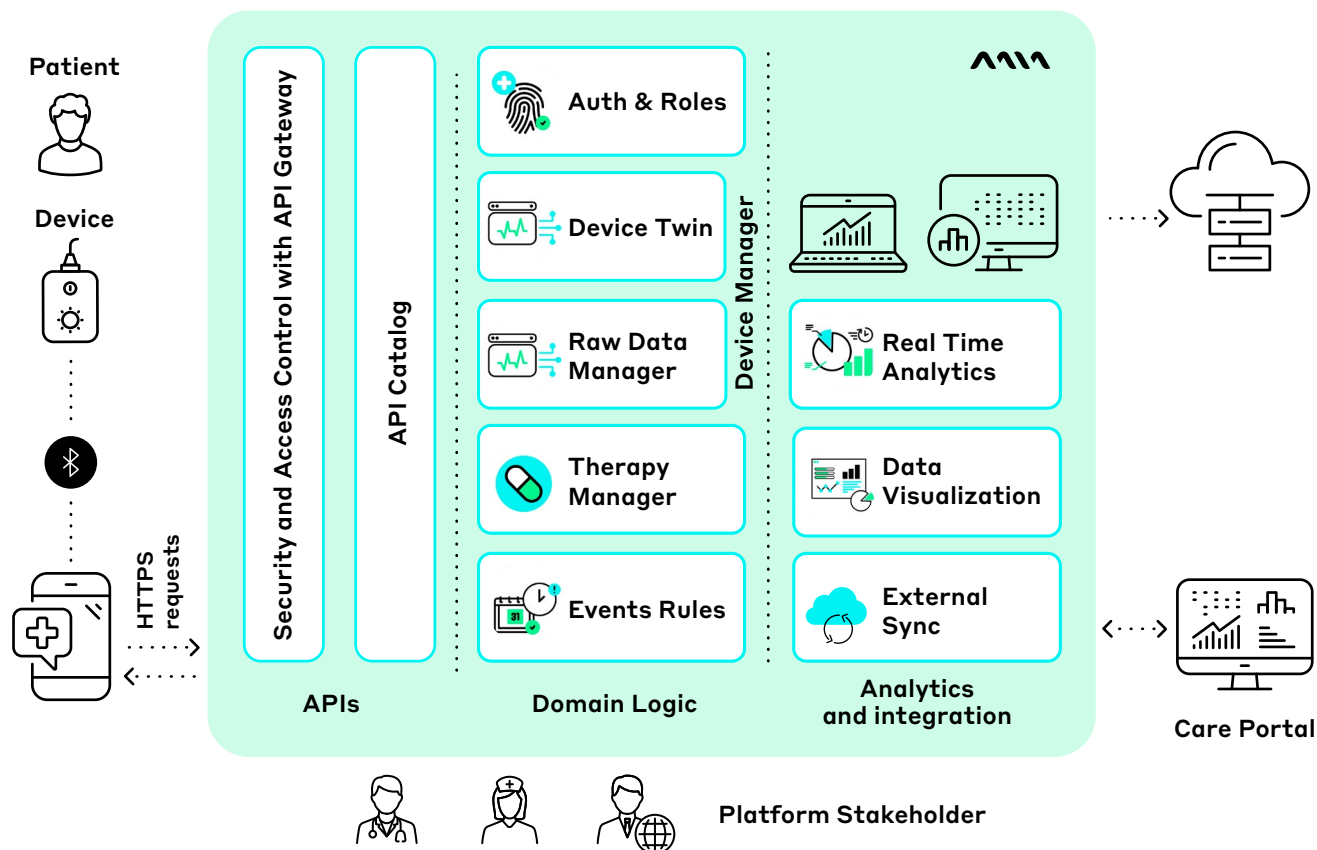
Develop Securely

Transparent and effective governance, with centralized monitoring of best practices for software development and the product life cycle

Connected Medical Devices: enhance interoperability and evolvability of services with Mia-Care

The platform developed by Mia-Care receives and processes data from medical devices, making them available to patients through a Mobile Application and to the Medical Staff through a monitoring dashboard (Desktop or Mobile) according to different authorization levels.

On the one end, the Medical equipe can monitor the correct use of the device and the adherence to therapy through the Care Portal, sending notifications and alerts to the patient monitored, and analyzing the treatment outcome for any further adjustments. On the other hand, the patient can use the Mobile Application to monitor the therapy, receive notifications from the caregiver and medical staff, request assistance, access safe and certified sources of medical information.



The system is composed by the following sections:

1. Domain Logics

Microservices developed on the needs and business logics characteristics of the MedTech.

2. Analysis and Integration

A layer consisting of several tools for analyzing the collected data and integrating with external systems.

3. Api Catalog

A safe and secure interface for the development of APIs to be exposed to external touchpoints and to the monitoring devices used by patients.

4. Api Gateway

A network layer that allows communication between the monitoring tools and the Cloud server, ensuring secure and controlled access to relevant data, based on specific authorization levels for each type of user.

To understand the value of the platform built on the needs of a MedTech company is necessary to get into the details of every specific component.

3.1

Domain Logics: the technological engine of the Platform that offers security and flexibility

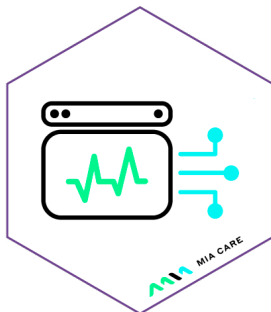
The microservices that compose the platform are both standard and customized. Still, for the management of connected medical devices, the platform is composed of ordinary logic, valid for all projects, and customized additional components that enable different functions tailored to the needs of the MedTech.

> Of course, given the evolutionary nature of Mia-Care technology, it is possible to **enrich the technology stack** by integrating additional digital applications for ancillary purposes.



Auth & Roles

Allows the creation of different authorization levels for each user and guarantees adequate data protection and security levels.



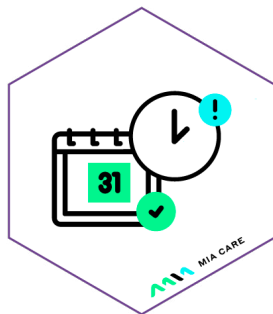
Device Manager

Allows you to remotely control the medical device, verify the correct functioning and correct configuration of the device, battery level, actual use, and update status (updating it to the latest release, remotely).



Therapy manager

It defines the configuration of different therapeutic paths based on the patient's needs and the state of maturity of the therapy followed. Thus, Medical Staff can pre-configure different therapeutic settings and associate them with a specific device based on the treatment prescribed to the patient.



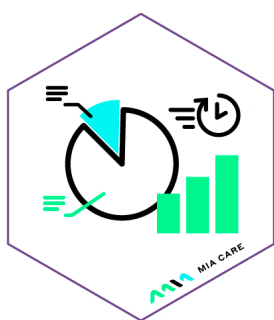
Events rules

Once the data has been transmitted and processed by the platform, it is possible to configure rules that permit the proactive management of the patient's behavior. For example, suppose the patient does not adequately follow the therapy. In that case, it is possible to set the sending of notifications on the associated smartphone App to stimulate him to respect the assigned treatments.

3.2

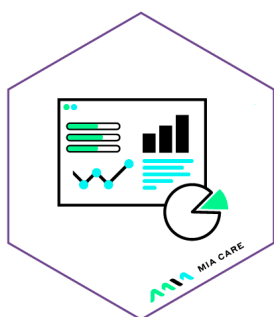
Analysis and Integration: the central component to foster clear governance of devices and the precise monitoring of patient therapy

This software layer enables you to gather information from storage systems connected to medical devices allowing healthcare companies to configure the data flow as needed. For instance, suppose the MedTech company can aggregate data according to business logics useful for product analysis. In this case, healthcare partners can take advantage of collecting information for clinical evaluations and therapeutic assistance.



Real Time Analytics

Through real-time analysis, the doctor can monitor the data transmitted through the Care Portal, verifying the correct adherence of therapy to assigned patients. At the same time, the MedTech manufacturer can collect valuable data on the clinical effectiveness of the medical device. Furthermore, this system contains helpful information on performance, use, distribution by geographical areas and demographic data, and devices' status (active, failures, deactivated).



Data Visualization

This digital tool allows users of Mia-Care technology (MedTech company, Specialist Doctors, and Administration Unit) to collect and organize all the information needed in a single view, according to the rules defined during the architecture design.

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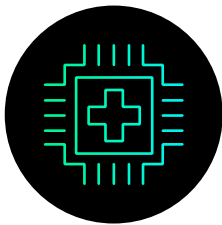
Conclusions

Adopting the platform to enhance the IoT activities of a MedTech led to a deep interchange of information within the stakeholders operating in the health space . It also led to a higher quality in terms of effectiveness of medical devices's use and adherence to therapy by patients.



An experience that has brought **tangible and significant benefits to the ecosystem** built around the Mia-Care platform.

The advantages for:



MedTech

- Customized, reliable, and regulatory compliant platform
- Lower development and implementation costs
- Time-to-market reduction for new services delivery
- Personalization of the architecture based on custom business logics and objectives
- Integration of medical devices with any information system
- Data to support incremental product innovation processes
- Evaluation of the clinical efficacy of the device on a large scale
- Remote verification of the performance and correct functioning of the devices
- Clear and structured governance, without bottlenecks on compliance and security



Clinics & Hospitals

- Greater availability and reliability of the services
- Easier and more convenient integration
- Remote evaluation of the treatment effectiveness and patient response
- Personalization of the therapeutic path
- Real-time verification of adherence to therapy
- Higher Patient Empowerment to follow the planned treatments
- More immediate doctor-to-patient communication
- Evaluation of the clinical effectiveness of the device through Analytics
- Improvement of Patient Engagement and Patient Empowerment



Patient

- Direct communication channel with specialists and caregivers
- Greater involvement and proactivity in the Therapy Management
- Facilitated adherence to therapy
- Remotization of Care
- Save time and resources thanks to fewer accesses to facilities
- Greater well-being and better quality of life
- Strengthening of Patient Empowerment



Mia-Care is an innovative startup that supports the digital transformation of **Pharma, Med Tech and Digital Health** companies. Mia-Care software suite contains unique capabilities and resources to seamlessly design, build, deliver and orchestrate digital services leveraging a modern web architecture based on **fast data, process automation, containerization and advanced analytics.**

The product is compliant by design with security and privacy regulations for EU and US applications and allows easy integration with external infrastructures. Learn more at **www.mia-care.io**.